



OpenShift Inventory Export Tool

This tool collects inventory and configuration data from an OpenShift cluster to support subscription analysis and environment understanding.

Outputs are structured and suitable for multi-cluster analysis and SEAP engagements.

Requirements

- Python 3.6+ (Python 3.7+ recommended)
- Access to the `oc` CLI and appropriate RBAC to read cluster resources

What It Collects

- Node roles, capacity, and configuration
- Installed operator subscriptions
- Add-on evidence (ACS/ACM/ODF/Quay)
- Cluster version, deployment type, and platform type (OCP vs OKE)
- DR posture heuristic + Hot DR indicators (signals-only)
- OpenShift Virtualization (VM) presence or absence

How to Use

1) Recommended (Token-based):

Obtain a token from the OpenShift console:
(User menu) -> Copy login command -> Display token
Then run:

```
python3 ocp_inventory_cli_v3.py --server https://api.<cluster>:6443 --token <sha256~...>
```

2) Username/password (only if supported by the cluster's OAuth provider):

```
python3 ocp_inventory_cli_v3.py --server https://api.<cluster>:6443 --username <user>
```

You will be prompted for the password.

3) Already logged in (SSO):

If `'oc whoami'` succeeds AND your current context server matches `--server`, the script will skip login automatically. Once logged in, the script will gather inventory data and create a zipped output folder automatically.

Run Modes

Default:

- Fast, low-risk, avoids large workload pulls
- Produces add-on evidence report and DR signals by default



Deep Mode:

- Adds cluster-wide workload evidence (pods) and a richer workload report
- May take longer on large clusters

```
python3 ocp_inventory_cli_v3.py --server https://api.<cluster>:6443 --token <sha256~...> --mode deep
```

Full Mode:

- Adds raw JSON dumps for review

```
python3 ocp_inventory_cli_v3.py --server https://api.<cluster>:6443 --token <sha256~...> --mode full
```

Outputs

Core Outputs:

- OCP_Nodes.csv
- OCP_Operators.csv
- OCP_Cluster_Version.txt
- OCP_Addons_Report.txt

Summary & Integrity:

- collection_summary.csv
- summary.json
- hash.txt

Optional Outputs:

- workload_evidence.csv (deep/full)
- raw_*.json (full)

Disconnected Environments

- Fully Supported
- No internet access required

Bare-Metal Guidance

For bare-metal deployments where socket-based subscription validation is required, additional verification may be needed. Physical CPU details can be captured using:

```
oc debug node/<node-name> -- chroot /host lscpu | grep -E 'Socket|Core'
```